P8109 Spring 2026 Homework #3

Due on April 9 at 11:59pm

1. Let X_1 and X_2 be two independent Poisson random variables with parameter λ. Let

$$T = X_1 + X_2.$$

   (a) By finding an expression for

$$P(X_1 = 1, X_2 = 2 \mid T = 3),$$

   show that T is not sufficient for λ.

   (b) By using the factorization theorem, obtain a sufficient statistic for λ.

2. Consider a discrete random variable X with probability mass function $p(x;\theta)$, where $x = 1, 2, 3$ and $\theta = \theta_1, \theta_2, \theta_3$. The values of $p(x;\theta)$ are given in the table below:

x	θ_1	θ_2	θ_3
1	0.1	0.2	0.3
2	0.7	0.4	0.1
3	0.2	0.4	0.6

Consider the statistic

$$T = \begin{cases} 0, & \text{if } x \text{ is odd}, \\ 1, & \text{if } x \text{ is even}. \end{cases}$$

   (a) Obtain the conditional distribution $p(x \mid T; \theta)$ in tabular form for $T = 0$ and for $T = 1$.

   (b) Comment on the nature of the statistic T.

3. Let $X_1, \ldots, X_n$ be i.i.d. random variables from the following PDF:

$$f(x;\theta) = \begin{cases} \theta^{-1}, & |\theta| < x < 2\theta, \\ 0, & \text{elsewhere}. \end{cases}$$

Obtain a sufficient statistic for θ.

4. Let X_1, X_2 be i.i.d. $N(\theta, 1)$. Let

$$d = (X_1 + X_2)/2, \qquad d^* = E(d \mid X_1).$$

   (a) Obtain the expectations of d and d^*.

(b) Show that

$$\mathrm{Var}(d^*) < \mathrm{Var}(d).$$

(c) Explain why d^* cannot be used as an improved estimator of θ.

5. Show that the Uniform$(\theta, \theta + 1)$ family is not complete.

6. Let $X_1, \ldots, X_n$ be i.i.d. random variables, each with a Binomial(m, p) distribution. We wish to obtain the UMVUE of

$p^r q^s$, where $q = 1 - p$ and r, s are known positive integers such that $r + s < mn$.

(a) By using moment-generating functions or otherwise, obtain the distribution of

$$T = X_1 + \cdots + X_n.$$

(b) Let $U(T)$ be an unbiased estimator of $p^r q^s$. By solving

$$E[U(T)] = p^r q^s,$$

show that

$$U(T) = \begin{cases} \dfrac{\binom{mn-s-r}{T-r}}{\binom{mn}{T}}, & T = r, r+1, \ldots, mn - s, \\ 0, & \text{otherwise.} \end{cases}$$

(c) By considering the family of distributions, argue briefly that $U(T)$ is the UMVUE of $p^r q^s$.